Bonds

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DATE PLACEHOLDER

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1 Bonds and Intermolecular Forces

1.1 Bond Types

Properties of matter depend on the types of bonds and intermolecular forces (IMF) present. **Covalent bonds** involve the sharing of electrons between atoms. **Ionic bonds** result from the electrostatic attraction between oppositely charged ions. **Metallic bonds** are formed by the sharing of valence electrons. **IMFs**, such as hydrogen bonding, dipole-dipole interactions, and London dispersion forces, influence properties like boiling point, melting point, and solubility.

1. **Molecular Covalent** (e.g. H_2, Cl_2, H_2O, CH_4) – *soft liquids*

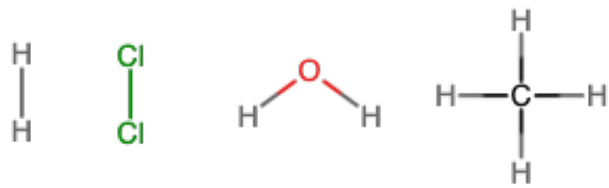

Figure 1: caption

2. **Network Covalent** (e.g. Carbon, Hydrocarbon, Sulfur) – *hard solids, non-brittle, non-conductive*

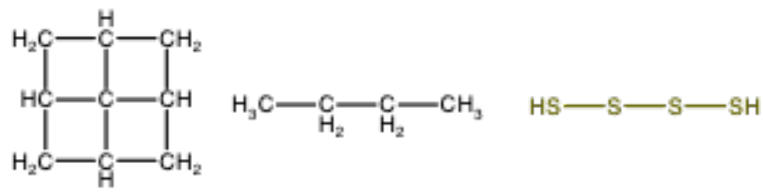

Figure 2: caption

3. **Ionic** (e.g. NaCl, $MgCl_2$, $CuSO_4$) – *anions(-) & cations(+) are held together by electrostatic forces*

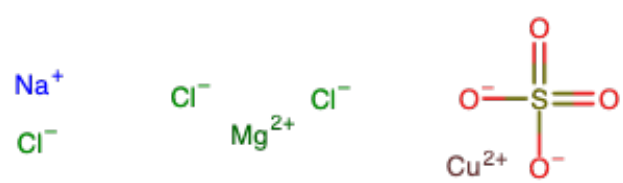

Figure 3: caption

4. **Metallic** (e.g. Li, Na, Ca, Al) – *malleable, bendable, conductor*

Metals consist of positively charged ions surrounded by a "sea of delocalized e^-s", allowing for high electrical and thermal conductivity, malleability, and ductility.

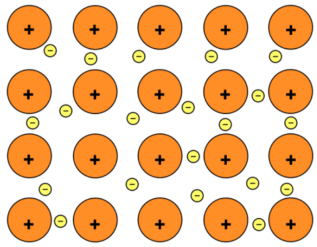

Figure 4: caption

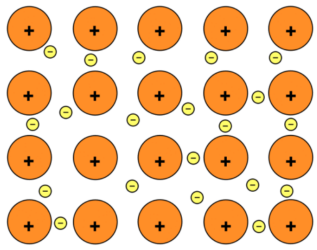

Figure 5: caption

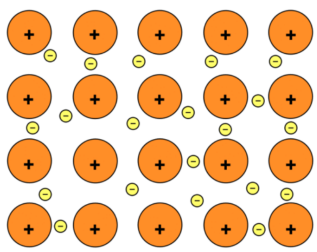

Figure 6: caption

1.2 VSEPR and Molecular Geometry

The shapes of molecules can be predicted using the **VSEPR (Valence Shell Electron Pair Repulsion)** theory, which states that electron pairs around a central atom will arrange themselves to minimize repulsion.

Number of Electron Dense Areas	Electron-Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2	Linear	Linear				
3	Trigonal planar	Trigonal planar	Bent			
4	Tetrahedral	Tetrahedral	Trigonal pyramidal	Bent		
5	Trigonal bipyramidal	Trigonal bipyramidal	Sawhorse	T-shaped	Linear	
6	Octahedral	Octahedral	Square pyramidal	Square planar	T-shaped	Linear

Figure 7: caption

CH_4 → no lone pairs, 4 e^- dense areas → **tetrahedral**
H_2O → 2 lone pairs, 4 e^- dense areas → **bent**
NH_3 → 1 lone pair, 4 e^- dense areas→ **pyramidal**.

Definition 1.1:

It is important to note here that **lone pairs take up more space than bonding pairs**, which can affect bond angles. For example, in NH_3, the bond angle is 107° instead of the ideal tetrahedral angle of 109.5° due to the presence of a lone pair on nitrogen. Similarly, in H_2O, the bond angle is 104.5° because of the two lone pairs on oxygen.

1.3 Bond Polarity and Electronegativity

The bond polarity of a molecule is determined by the difference in **electronegativity (EN)** between the bonded atoms. The strength of the bonds from weakest to strongest can be categorized as follows:

1. **Nonpolar Covalent**: EN difference < 0.5 – (*e.g.* H_2, Cl_2, F_2)

2. **Polar Covalent**: EN difference between 0.5 and 1.7 –(*e.g.* H_2O, NH_3, HCl)

3. **Ionic**: EN difference > 1.7 – (*e.g.* $NaCl$, MgO, CaF_2)

The EN values of elements are fundamentally determined by the laws of electromagnetic forces, which depend on the **strength of opposing charges** (Q_1 & Q_2) and the **distance between them** (r). This is described by Coulomb's Law:

$$F = k\frac{Q_1 Q_2}{r^2}$$

As you move across the periodic table from left to right (→), EN increases due to the increasing number of p^+ attracting the same number of e^- shells. **As you move downwards (↓), EN decreases** because the atomic radius increases, reducing the effective nuclear charge (Z_{eff}) felt by the valence electrons.

The Z_{eff} can be approximated using the formula

$$Z_e ff = Z - S$$

where Z is the actual nuclear charge (number of p^+) and S is the number of e^- within the valence shell (*this excludes the valence e^-s*).

Definition 1.2:

Keep in mind that **the actual definition of Z_{eff} is more nuanced** than the formula above. The shielding effect of inner electrons is not uniform, and different orbitals (s, p, d, f) have varying effects on Z_{eff}. For example, s-electrons are more effective at shielding than p-electrons. Additionally, electron-electron repulsions within the same shell can also affect the effective nuclear charge experienced by valence electrons.

1.4 Intermolecular Forces

Before discussing intermolecular forces (IMFs), it is important to understand that although an individual bond may be polar, the **overall molecule can be nonpolar if the molecular geometry allows for the dipole moments to cancel out**. For example, CO_2 has polar bonds, but the linear geometry results in a nonpolar molecule.

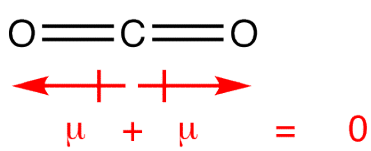

Figure 8: caption

Thus, when evaluating molecular polarity, both bond polarity and molecular geometry must be considered.

IMFs are the forces of attraction/repulsion **between** molecules. They are generally weaker than covalent or ionic bonds but play a crucial role in determining the physical properties of substances. The main types of IMFs include:

1. **London Dispersion Forces (LDFs)** – present in all molecules, but dominant in nonpolar molecules. They arise from temporary dipoles created by the movement of e^-. The strength of LDFs increases with the size and shape of the molecule (*more surface area = stronger LDFs*).

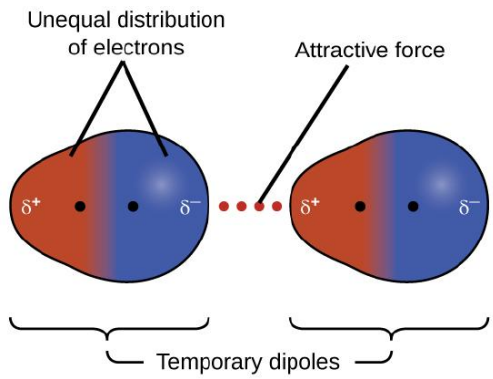

Figure 9: caption

Definition 1.3:

The boiling point (BP) of alkanes (C_xH_y) increases with increasing molar mass due to stronger LDFs. Similarly, a larger surface area in molecules like long-chain hydrocarbons leads to higher BPs compared to their more compact isomers due to an increased area for LDF interactions.

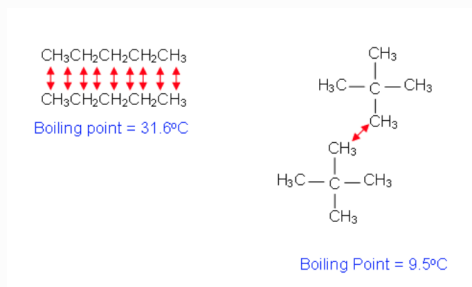

Figure 10: caption

2. **Dipole-induced Dipole & Ion-induced Dipole** – occurs when a polar molecule or ion induces a dipole in a nonpolar molecule by distorting its electron cloud.

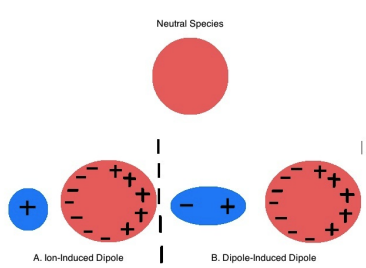

Figure 11: caption

3. **Dipole-Dipole Interactions** – occur between polar molecules where the (+) end of one molecule is attracted to the (-) end of another.

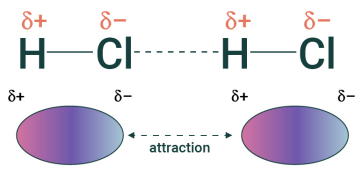

Figure 12: caption

4. **Hydrogen Bonding** – a special case of dipole-dipole interaction that occurs when H is bonded to highly electronegative atoms like N, O, or F.

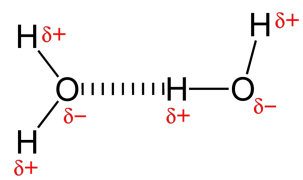

Figure 13: caption

4. **Ion-Dipole Interactions** – occur between ionic compounds and polar molecules, such as when salt (NaCl) dissolves in water.

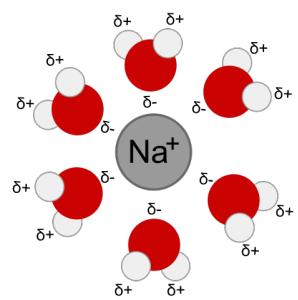

Figure 14: caption

Definition 1.4:

SIDE NOTE: **Formal Charge (FC)** is a useful concept for determining the most stable Lewis structure for a molecule. It is calculated using the formula

$$FC = V - (L + \frac{B}{2})$$

V → # of valence e^-
L → # of lone pairs
B → # of bonding e^-

The **most stable Lewis structure typically has the smallest formal charges** on atoms, with negative formal charges residing on the more electronegative atoms.

For example, in NO_3^-, N has a formal charge of +1, while one of the O atoms has a formal charge of -1, and the other two O atoms have a formal charge of 0. This distribution of formal charges helps to explain the resonance structures and overall stability of the nitrate ion.